TOPIC 5.4

# Bias–Variance Tradeoff

Why simple models miss the pattern and flexible models chase the noise

**WHAT YOU WILL LEARN**

- Define **bias**, **variance** and **irreducible error** in plain words
- Connect **high bias to underfitting** and **high variance to overfitting**
- Explain the decomposition **Total error = Bias² + Variance + Irreducible error**
- Describe the **tradeoff** and how to find a good balance
- See each term measured in a simulation

## 1. The idea in one example

Suppose we predict **house price from size (square feet)**. We draw a **straight line** through some houses. Real prices curve a little, so a straight line will be **consistently off**, no matter which houses we sampled. That consistent miss is **bias**.

Now suppose we use a very wiggly curve that passes through every house. It fits our sample perfectly, but if we had collected a *different* sample of houses, the wiggles would land somewhere else and the predictions would change a lot. That sensitivity to the particular sample is **variance**.

| | Bias | Variance |
|---|---|---|
| Meaning | How far the model's **average** prediction is from the truth | How much the prediction **changes** when the training data changes |
| Cause | A model that is too simple, with wrong assumptions | A model that is too flexible and reacts to noise |
| Leads to | **Underfitting** | **Overfitting** |
| House example | A straight line for a curved relationship | A wiggly curve through every house |

## 2. The formula

Imagine training the same kind of model on many different samples and predicting the same new point each time. The expected squared error splits into three parts:

$$\text{Total error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible error}$$

- **Bias²**: (average prediction − true value)². The systematic error.
- **Variance**: how much predictions spread around their own average.
- **Irreducible error**: noise in the data itself (measurement error, things we did not record). **No model can remove it.**

## 3. Seeing it: many fits of the same model

To measure these terms directly, we use a known true curve (a smooth wave, dashed) and repeat this 300 times: draw a fresh sample of 30 noisy points, fit the model, and keep the predictions. Figure 1 shows 25 of those fits (blue) and their average (orange).

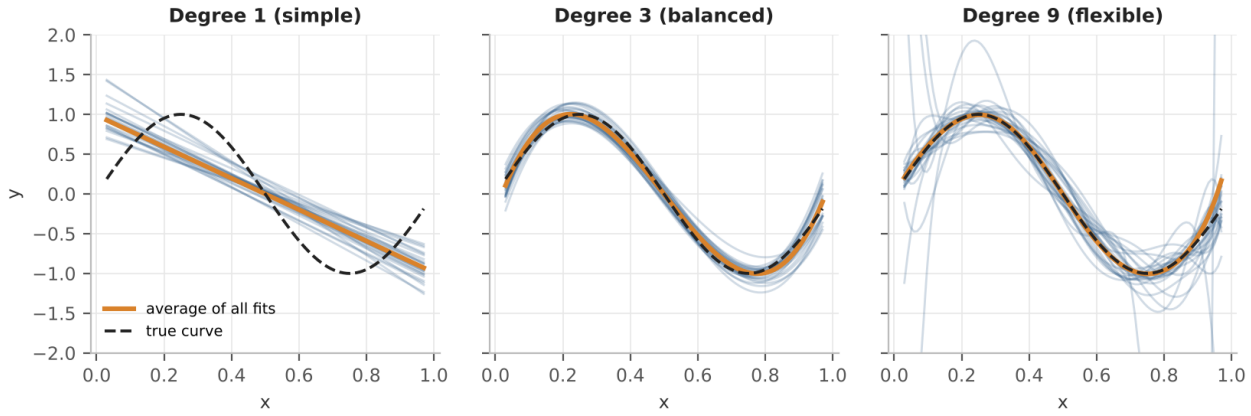



Figure 1. Each blue line is the same model trained on a different sample. The orange line is their average.

- **Degree 1:** the blue lines are tightly bunched (low variance) but all far from the true curve, and the average is a straight line that misses the wave (**high bias**).
- **Degree 3:** the lines follow the wave and stay fairly close together: low bias, low variance.
- **Degree 9:** on average the fits follow the wave, so bias is small, but the individual lines scatter widely (**high variance**).

## 4. Measured numbers

| Degree | Bias² | Variance | Noise² | Total error |
|---|---|---|---|---|
| 1 | 0.165 | 0.017 | 0.062 | 0.245 |
| 3 | 0.003 | 0.010 | 0.062 | 0.076 |
| 5 | 0.000 | 0.024 | 0.062 | 0.086 |
| 7 | 0.000 | 0.140 | 0.062 | 0.203 |
| 9 | 0.003 | 2.413 | 0.062 | 2.489 |

Bias² collapses from 0.165 (degree 1) to 0.003 (degree 3) and stays near zero. Variance, small for simple models, rises steeply, from 0.010 at degree 3 to 2.413 at degree 9. The total error is lowest at degree 3: the balance point. The noise term (0.062, which is $0.25^2$) never changes.

The total in the last column is measured against fresh noisy values, so it matches Bias² + Variance + Noise² up to small random differences from the simulation.

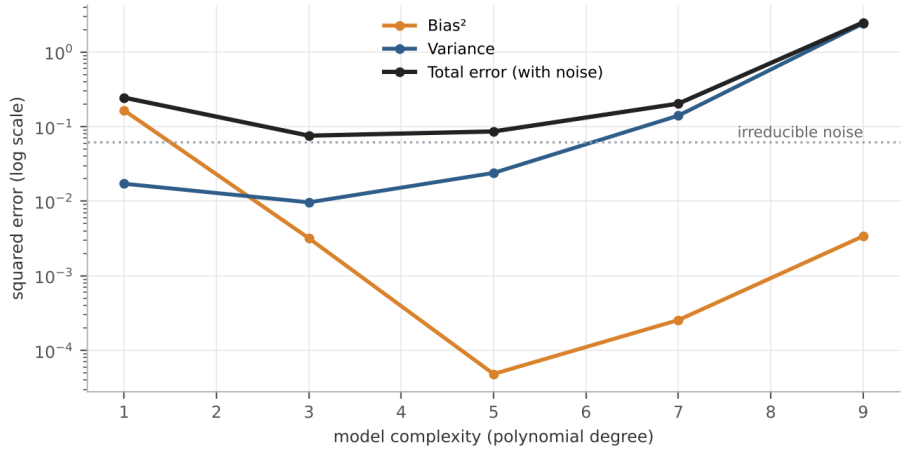



Figure 2. Bias² drops quickly, then variance takes over. The total error is lowest in between. (Log scale: each gridline is ten times the one below.)

## 5. The tradeoff

You cannot usually lower bias and variance together by changing complexity alone: **reducing one tends to raise the other**. The goal is the point where their sum is smallest.

| | High bias (too simple) | High variance (too complex) |
|---|---|---|
| Training error | High | Very low |
| Test error | High | High (much above training) |
| Typical fix | More flexible model, better features | More data, simpler model, regularization |

> **NOTE** **Ways to improve the balance**
>
> - **More training data** lowers variance without raising bias
> - **Regularization** (Ridge, Lasso: see 5.5) deliberately accepts a little bias to cut a lot of variance
> - **Cross-validation** (5.3) helps choose the complexity where test error is lowest
> - **Ensembles** such as bagging and random forests average many high-variance models to reduce variance

## Summary

> **SUMMARY**
>
> - **Bias**: systematic error from a model that is too simple (underfitting)
> - **Variance**: sensitivity to the training sample from a model that is too flexible (overfitting)
> - **Total error = Bias² + Variance + Irreducible error**; the last part cannot be removed
> - As complexity rises, **bias falls and variance rises**; aim for the minimum of the total
> - More data and regularization are the main tools to improve the balance

## Practice questions

Try each question before reading the answer underneath it.

**Q1.** A model has high bias. What happens to its training and test error?

**Answer.** Both are **high**: the model is too simple to capture the pattern (underfitting).

**Q2.** What is irreducible error?

**Answer.** Random noise in the data itself. Even a perfect model cannot predict it, so it sets a floor on the error.

**Q3.** Why does a very flexible model have high variance?

**Answer.** It follows the noise in whichever sample it was trained on, so its predictions change a lot from one sample to another.

**Q4.** Write the decomposition of the total expected error.

**Answer.** Total error = Bias² + Variance + Irreducible error.

**Q5.** How does regularization help?

**Answer.** It restricts the model so its predictions vary less between samples. This costs a little extra bias but usually lowers the total error.